

Master Thesis
BlueRov2 Formation Control

Week 6

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Next Steps

- Formation control with single integrator agents no constraints
  - Undirected graph
  - Directed graph
- Single integrator formation control with constraints
  - Only connectivity maintenance
  - Then also colision avoidance
- Double integrator agents



- BlueRov2 model no input constraints
  -
- [Research challenge] integrate input constraints
  -
- [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)

Achievements

- Study
  - Graph Theoretic Methods in Multiagent Networks
  - Esteban’s thesis
  - Khalil's book on Nonlinear Control

- Implementation
  - Improvements
    - Generalized and modularized the code
    - Added Formation Offset error
    - Tuned and changed variables into more realistic ones
  - Implemented Field of View constraint

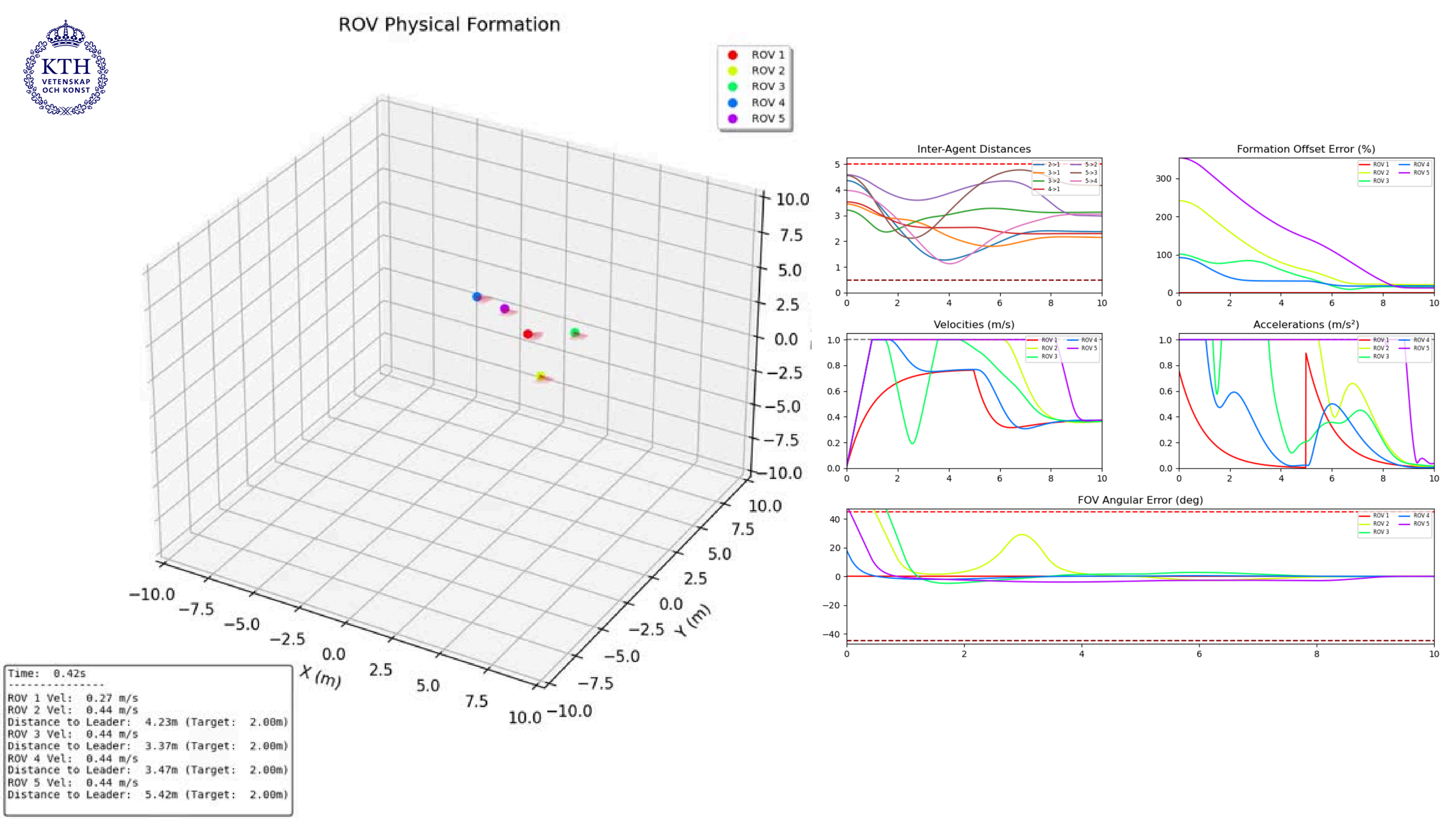
Command-filtered backstepping

$$W_{\alpha,k}(\alpha_k) = \frac{1}{2}\left[\alpha_k^2 + \ln\left(\frac{\Delta_{\alpha,k}^2}{\Delta_{\alpha,k}^2 - \alpha_k^2}\right)\right]$$

$$\nabla W(\alpha_k) = \left(\frac{1}{\Delta_{\alpha,k}^2 - \alpha_k^2}\right)\alpha_k$$

$$\omega_i = -k_p \alpha_{error} - c \sum_{K \in \mathcal{N}_i} \nabla W(\alpha_k)$$

$$\alpha_{error} = \psi_i - \psi_{target}$$


ROV Physical Formation
ROV 1
ROV 2
ROV 3
ROV 4
ROV 5
X (m)
Y (m)
Inter-Agent Distances
Formation Offset Error (%)
Velocities (m/s)
Accelerations (m/s²)
FOV Angular Error (deg)
Time: 0.42s
ROV 1 Vel: 0.27 m/s
ROV 2 Vel: 0.44 m/s
Distance to Leader: 4.23m (Target: 2.00m)
ROV 3 Vel: 0.44 m/s
Distance to Leader: 3.37m (Target: 2.00m)
ROV 4 Vel: 0.44 m/s
Distance to Leader: 3.47m (Target: 2.00m)
ROV 5 Vel: 0.44 m/s
Distance to Leader: 5.42m (Target: 2.00m)

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Field of View

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Questions & Comments